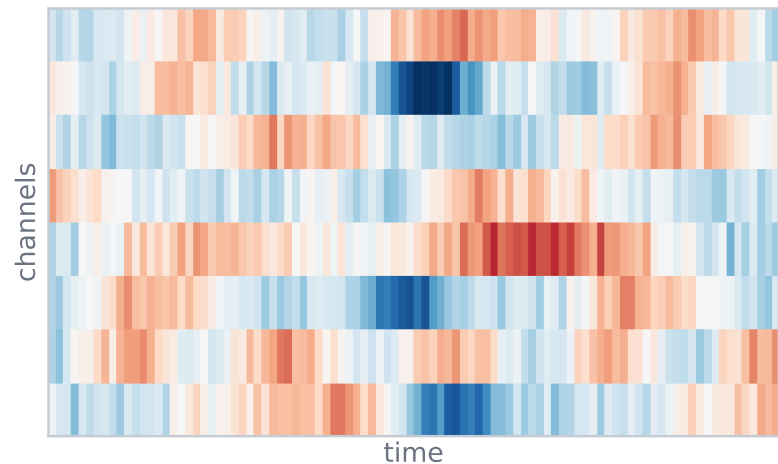


# NFPA inductive bias: separable channel-time structure

## A EMPCA basis vector $p_i$

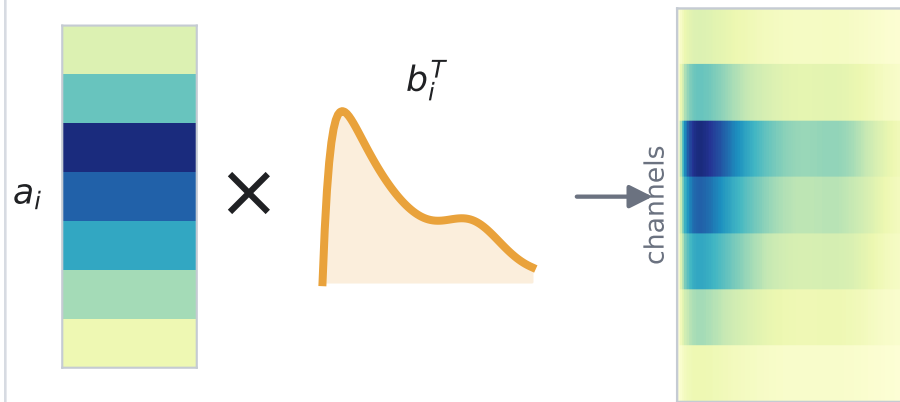
arbitrary channel-time pattern



one free pattern across all channels and all time samples

## B NFPA factorization

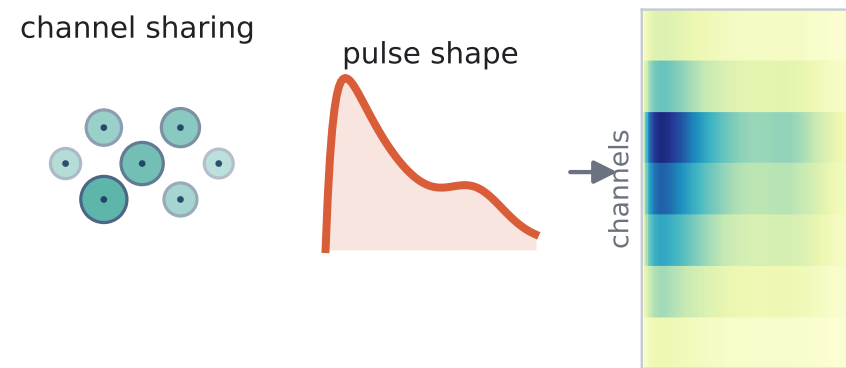
basis constrained to be separable



**NFPA basis = channel mode  $\times$  time mode**

## C Detector interpretation

spatial sharing and pulse evolution



**NFPA learns spatial sharing  $\times$  pulse evolution**